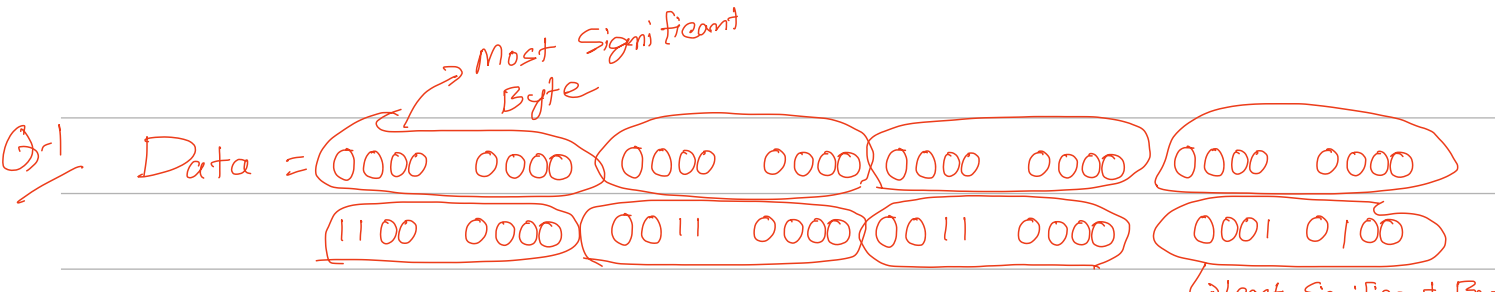


Q-1 Data = 

- How many bits can you store in 1 memory location/cell?
- Show how the given Data will be stored in memory of a RISC-V machine. Assume the starting memory location is 0
- Show how the given Data will be stored in memory of a machine that follows big endian. Assume the starting memory location is 8 approach

Solution—

a) 8 bits.

b) RISC-V follows little endian approach.

Data Size = 64 bits. So, Memory cells required = $\frac{64}{8} = 8$ cells.

Starting location = 0.

Data Memory

0	0001 0100 → LSB
1	0011 0000
2	0011 0000
3	1100 0000
4	0000 0000
5	0000 0000
6	0000 0000
7	0000 0000 → MSB

c) Data Size = 64 bits. So, Memory cells required = $\frac{64}{8} = 8$ cells.
Starting location = 8.

Data Memory	
8	0000 0000 → MSB
9	0000 0000
10	0000 0000
11	0000 0000
12	1100 0000
13	0011 0000
14	0011 0000
15	0001 0100 → LSB

↙ Big Endian

Q-2 Data = (AB13579CD)₁₆

a) How many bits can you store in 1 memory location/cell?

b) Show how the given Data will be stored in memory of a risc-v machine. Assume the starting memory location is 0.

c) Show how the given Data will be stored in memory of a machine that follows big endian approach. Assume the starting memory location is 12.

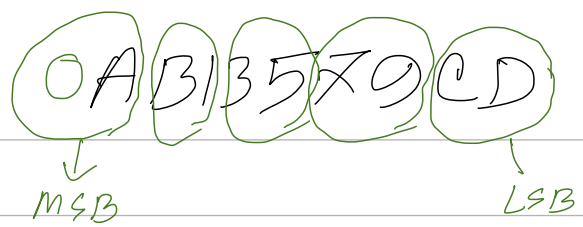
Solution -

a) 8 bits.

b) RISC-V follows little endian approach.

Data Size = 36 bits. So, Memory cells required = $\frac{36}{8} = 4.5$
Starting location = 0. = 5

Data Memory	
0	CD → LSB
1	79
2	35
3	B1
4	0A → MSB

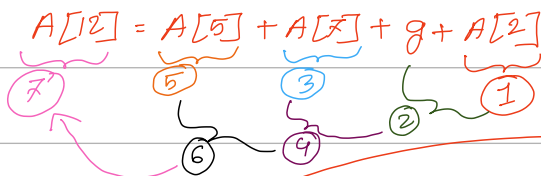


① Data Size = 36 bits. So, memory cells required = $\frac{36}{8} = 4.5$ cells.
Starting location = 12. = 5 4

Data Memory	
12	0A → MSB
13	B1
14	35
15	79
16	CD → LSB

↙ Big Endian

Q-3 Array A is a Byte array with size of 50. Base address of A is stored in X_{23} . Variable g & h are associated with X_5, X_6 registers. Write the necessary risc-V code of the following code.



Sol:

LB $X_7, 2(X_{23})$; $X_7 = A[2] \Rightarrow 2 \times 1$
 Add X_7, X_5, X_7 ; $X_7 = X_5 + X_7$
 LB $X_8, 7(X_{23})$; $X_8 = A[7] \Rightarrow 7 \times 1$
 Add X_8, X_7, X_8 ; $X_8 = X_7 + X_8$
 LB $X_9, 5(X_{23})$; $X_9 = A[5] \Rightarrow 5 \times 1$
 Add X_9, X_8, X_9 ; $X_9 = X_8 + X_9$
 SB $X_9, 12(X_{23})$; Value of $X_{23} + 12$ gives the mem location of $A[12]$. Storing the value of X_9 in that memory location.

Q-4 Data Memory: (little endian)

Location	Data	Loc.	Data	Loc.	Data	Loc.	Data	$X_{21} = (12)_{10}$
21	$(AC)_{16}$	✓24	$(D0)_{16}$	✓27	$(00)_{16}$	✓30	$(00)_{16}$	$X_{22} = (13)_{10}$
22	$(12)_{16}$	✓25	$(FE)_{16}$	✓28	$(00)_{16}$	31	$(F9)_{16}$	
✓23	$(05)_{16}$	✓26	$(23)_{16}$	✓29	$(00)_{16}$	32	$(23)_{16}$	

Instruction:

LD $X_5, 10(X_{22})$

After executing the above instruction, what is the value of X_5 ?

Sol:

$X_{22} = 13$; Mem location = Value of the Base + Offset
register

$$= 13 + 10$$

$$= 23 \leftarrow \text{Starting location of Data.}$$

LD → Double Word → working with 64 bits data.

So, 8 continuous cells we need.

Location	Data
23	$(05)_{16} \rightarrow \text{LSB}$; Data stored using little endian approach.
24	$(D0)_{16}$
25	$(FE)_{16}$
26	$(23)_{16}$
27	$(00)_{16}$
28	$(00)_{16}$
29	$(00)_{16}$
30	$(00)_{16} \rightarrow \text{MSB}$

$$\text{Data} = (00\ 00\ 00\ 00\ 23\ FE\ D0\ 05)_{16}$$

$$X_5 = (00000000\ 23\ FE\ D0\ 05)_{16}$$

Q-5 Data Memory: (little endian)

Location	Data	Loc.	Data	Loc.	Data	Loc.	Data	$X_{21} = (12)$
21	0011 1011	24	0000 1111	27	1111 1111	30	0011 0000	$X_{22} = (13)$
22	0011 0000	25	1011 0000	28	1011 0011	31	1000 0001	
23	0001 1111	26	0000 1011	29	0001 1000	32	1000 1111	

Instruction: $SB \ X_5, 10(X_{22})$

$$\begin{array}{cccccccc}
 \text{msb} & & & & & & & \\
 0000 & 0000 & 0000 & 0000 & 0000 & 0000 & 0000 & 0000 \\
 X_5 = & 0000 & 0000 & 0000 & 0000 & 0000 & 0000 & 1011 \ 1010 \ \text{lsb}
 \end{array}$$

After executing the above instruction, update the data memory.

Sol:

$X_{22} = 13$; Mem location = Value of the Base + Offset
register

$$= 13 + 10$$

$= 23 \leftarrow$ Starting location of Data.

$S(B) \rightarrow$ Byte $\rightarrow$ working with 8 bits data.

So, 1 cell is needed.

$$\begin{array}{cccccccc}
 \text{msb} & & & & & & & \\
 0000 & 0000 & 0000 & 0000 & 0000 & 0000 & 0000 & 0000 \\
 X_5 = & 0000 & 0000 & 0000 & 0000 & 0000 & 0000 & 1011 \ 1010 \ \text{lsb}
 \end{array}$$

8 bit

Location	Data	Loc.	Data	Loc.	Data	Loc.	Data
21	0011 1011	24	0000 1111	27	1111 1111	30	0011 0000
22	0011 0000	25	1011 0000	28	1011 0011	31	1000 0001
23	0001 1111 1011 1010	26	0000 1011	29	0001 1000	32	1000 1111